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Can Fine-Tuned LLMs Beat Knowledge Tracing? A Systematic Benchmark with Open-Source Models

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Abstract

Knowledge tracing (KT) — predicting whether a student will answer a question correctly given their learning history — is fundamental to adaptive learning systems that now serve millions of learners worldwide. While large language models (LLMs) have recently been applied to KT, existing comparisons rely on closed-source models with inconsistent evaluation protocols, yielding non-reproducible results and unreliable cost estimates. This study presents the first systematic benchmark comparing four established KT models (BKT, DKT, AKT, simpleKT), three open-source LLM approaches (BERT fine-tuned, LLaMA 3 8B with LoRA, Qwen 2.5 7B with LoRA), and a novel hybrid architecture (simpleKT backbone augmented with LLM-derived question embeddings) on two standardized datasets — ASSISTments 2009 and EdNet — under identical preprocessing and evaluation procedures via the pyKT framework. Our results reveal three key findings. First, specialized KT models (simpleKT, DKT) achieve strong baseline performance (AUC 0.81 on ASSISTments 2009) at sub-5 ms inference latency. Second, fine-tuned LLMs demonstrate a clear advantage in cold-start scenarios where students have fewer than ten recorded interactions, leveraging semantic question understanding to compensate for sparse behavioral data. Third, the proposed hybrid architecture achieves the best cost-performance tradeoff (AUC 0.828, 12 ms latency), outperforming both standalone paradigms while maintaining practical deployment efficiency. These findings provide actionable deployment guidance: KT models for latency-sensitive production systems, LLMs for cold-start enrichment, and the hybrid approach for platforms seeking optimal predictive accuracy within feasible computational budgets.

1 Introduction

Adaptive learning systems have transformed education at scale, serving millions of students through intelligent tutoring platforms that personalize instruction based on real-time assessments of learner knowledge (Corbett and Anderson 1995; Romero and Ventura 2020). At the core of these systems lies knowledge tracing (KT), the task of modeling a student's evolving mastery of knowledge components and predicting their performance on future exercises. Given a student's history of interactions — questions attempted, correctness of responses, and the knowledge components involved — a KT model estimates the probability that the student will answer the next question correctly. Modern educational platforms generate billions of learning interactions annually, and the accuracy of KT models directly determines the quality of content recommendations, pacing decisions, and intervention timing (Gligorea et al. 2023; Bond et al. 2024). Platforms such as ASSISTments, Khan Academy, and Duolingo depend on KT to decide when a student has mastered a skill and should advance, or when additional practice is needed. As educational institutions increasingly adopt AI-driven tools across both K-12 and higher education settings, the demand for accurate, efficient, and practically deployable KT solutions has never been greater (Alnasyan et al. 2024).

Despite three decades of progress — from Bayesian Knowledge Tracing (Corbett and Anderson 1995) to deep learning architectures such as DKT (Piech et al. 2015) and attention-based

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models like simpleKT (Liu et al. 2023a) — the KT landscape faces a new disruption. Large language models (LLMs) have demonstrated remarkable capabilities across diverse natural language processing tasks, and recent studies have begun exploring their application to student performance prediction (Dong et al. 2024; Guo et al. 2024; Kim et al. 2025). The appeal is intuitive: LLMs can leverage the semantic content of questions and answer options, potentially capturing pedagogical relationships that purely interaction-based KT models cannot access. Early results suggest that LLMs may hold particular advantages in cold-start scenarios, where new students have too few interactions for traditional models to form reliable estimates (Guo et al. 2024).

However, these emerging LLM-based KT approaches suffer from two critical limitations that undermine their scientific and practical value. First, existing comparisons overwhelmingly rely on closed-source models (GPT-3, GPT-4), producing results that are neither reproducible nor cost-applicable for educational institutions considering on-premise deployment. The emergence of capable open-source alternatives — LLaMA (Touvron et al. 2023), Qwen (Bai et al. 2023) — combined with parameter-efficient fine-tuning methods such as LoRA (Hu et al. 2022), fundamentally changes the accessibility equation, yet no study has evaluated these models for KT. Second, evaluation protocols remain fragmented: the pyKT benchmark (Liu et al. 2022) revealed that reported AUC scores for DKT on ASSISTments 2009 range from 0.721 to 0.821 across publications due to inconsistent preprocessing and splitting strategies (Naranjo and Stoffova 2025). When LLM approaches are evaluated against such moving baselines, it becomes impossible to draw reliable conclusions about their relative merit.

To address these gaps, this study makes three contributions:

1. **Reproducible benchmark.** We present the first systematic comparison of four established KT models (BKT, DKT, AKT, simpleKT), three open-source LLM approaches (fine-tuned BERT, LLaMA 3 8B with LoRA, Qwen 2.5 7B with LoRA), and a novel hybrid architecture on two standardized datasets — ASSISTments 2009 (Feng et al. 2009) and EdNet (Choi et al. 2020) — using identical preprocessing, five-fold cross-validation splits, and evaluation metrics within the pyKT framework (Liu et al. 2022).
2. **Comprehensive scenario analysis.** We conduct the first cost-performance-latency Pareto analysis that systematically compares KT and open-source LLM paradigms across both normal and cold-start scenarios (students with fewer than ten interactions), extending beyond the single-scenario evaluations of prior work (Guo et al. 2024; Kim et al. 2025).
3. **Hybrid architecture with practical deployment guidance.** We propose a lightweight hybrid model that augments a simpleKT backbone with LLM-derived question embeddings, achieving state-of-the-art predictive accuracy (AUC 0.828) at 10–50× lower inference cost than full LLM approaches, providing actionable recommendations for educational platforms operating under real-world computational constraints (Plooy et al. 2024).

The remainder of this paper is organized as follows. Section 2 reviews related work on knowledge tracing models, LLMs in educational AI, and recent LLM-based KT approaches. Section 3 describes our methodology, including datasets, model configurations, and evaluation protocols. Section 4 presents the experimental results across main performance, cold-start, and cost-performance analyses. Section 5 discusses the practical implications and limitations of our findings. Section 6 concludes with recommendations and future research directions.

2 Related Work

2.1 Knowledge Tracing Models

Knowledge tracing has evolved through three distinct generations of modeling approaches. The first generation established probabilistic foundations: Bayesian Knowledge Tracing (BKT)

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(Corbett and Anderson 1995) modeled student mastery as a hidden Markov process with learn and guess parameters, while Performance Factors Analysis (PFA) (Pavlik et al. 2009) introduced logistic regression over success and failure counts, enabling efficient model search and adaptive practice selection. These models remain widely deployed due to their interpretability and low computational cost, but their capacity to capture complex temporal learning dynamics is inherently limited.

The second generation leveraged deep learning to overcome these constraints. Deep Knowledge Tracing (DKT) (Piech et al. 2015) applied recurrent neural networks (LSTMs) to interaction sequences, demonstrating substantial improvements over BKT without requiring explicit knowledge component engineering. Dynamic Key-Value Memory Networks (DKVMN) (Zhang et al. 2017) introduced external memory to store and update concept-level mastery states explicitly, while the Self-Attentive Knowledge Tracing model (SAKT) (Pandey and Karypis 2019) pioneered the use of attention mechanisms for KT, addressing data sparsity through selective focus on relevant past interactions. Recent attention-based approaches include ELAKT (Pu et al. 2024), which enhances locality awareness in the attention mechanism to better capture short-range learning dependencies.

The current third generation refines attention-based architectures with psychometric grounding. Attentive Knowledge Tracing (AKT) (Ghosh et al. 2020) integrates Rasch-model-inspired embeddings with exponential decay attention to capture both question difficulty and forgetting dynamics. simpleKT (Liu et al. 2023a) demonstrated that explicit question-centric difficulty modeling achieves competitive performance with minimal architectural complexity, consistently ranking among the top three models across seven public datasets. sparseKT (Huang et al. 2023) further improved robustness by applying k-sparse attention to filter irrelevant learning events, reducing overfitting on small-scale educational datasets.

Despite this modeling progress, standardized benchmarking remains a persistent challenge. The pyKT toolkit (Liu et al. 2022) provided the first unified evaluation framework for deep learning-based KT models, revealing alarming inconsistencies in reported results. Naranjo and Stoffova (2025) extended this analysis, finding that evaluation protocol choices — including data splitting strategy, sequence length truncation, and negative sampling — account for more variance in reported performance than the choice of model architecture itself. These findings underscore the necessity of conducting all model comparisons under strictly identical experimental conditions, a principle we adopt throughout this study.

2.2 LLMs in Educational AI

Large language models have rapidly permeated educational technology applications beyond knowledge tracing. In intelligent tutoring systems, LLMs serve as conversational agents that provide explanations, hints, and scaffolded feedback (Cao 2023; Létourneau et al. 2025), substantially reducing the authoring effort required for traditional rule-based tutors. Automated essay scoring has similarly benefited from LLM capabilities, with recent studies demonstrating that fine-tuned models approach human-level reliability on standardized writing assessments (Pack et al. 2024). More broadly, LLMs have been applied to educational survey design and analysis, where Parker et al. (2024) demonstrated that language models can assist in constructing and validating assessment instruments at scale.

In learning analytics and student modeling, LLMs have been applied to predict at-risk students (Krüger et al. 2023; Alnasyan et al. 2024), generate personalized learning recommendations (Ni et al. 2024), and analyze learning behavior patterns at scale (Romero and Ventura 2020). Predictive models leveraging deep learning have also been explored for early identification of student dropout risk, with Almalawi et al. (2024) demonstrating the potential of neural network-based approaches for proactive intervention in online learning environments. Dong et al. (2024) provided a

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comprehensive systematic review of LLM applications across educational contexts, identifying student performance prediction as one of the most promising yet under-explored directions. Bond et al. (2024) further documented the accelerating adoption of AI in education, noting that the gap between research demonstrations and practical deployment remains substantial, particularly regarding computational cost and infrastructure requirements.

2.3 LLM-based Knowledge Tracing

The direct application of LLMs to knowledge tracing is an emerging research frontier with a small but rapidly growing body of work. Kim et al. (2025) proposed a text-option weighting approach that enables token-efficient LLM-based KT, demonstrating that incorporating question text and answer option information improves prediction accuracy. Their evaluation, however, relied on closed-source API models, limiting reproducibility and practical cost analysis. Guo et al. (2024) addressed the cold-start problem through attribute-aware LLM approaches, showing that LLMs can leverage question content and metadata to make reasonable predictions even for students with minimal interaction history — a scenario where traditional KT models typically fail due to insufficient behavioral data. Related work on concept prerequisite learning (Mazumder et al. 2023) has explored how language models can infer the dependency structure among knowledge components, a capability that could further enhance LLM-based KT by enabling explicit prerequisite-aware prediction.

The conceptual framing of KT as a next-token prediction task — where student interaction histories are serialized into text sequences and correctness predictions are generated autoregressively — represents a natural bridge between KT and language modeling (Song et al. 2022). This framing enables direct application of parameter-efficient fine-tuning methods such as LoRA (Hu et al. 2022) to adapt pre-trained language models for KT with modest computational overhead. However, existing implementations have not been systematically compared against the full spectrum of modern KT baselines under controlled conditions, nor have their cost-performance characteristics been rigorously quantified.

2.4 Summary of Research Gaps

Two critical gaps emerge from this literature review. First, no existing study benchmarks open-source LLMs (e.g., LLaMA (Touvron et al. 2023), Qwen (Bai et al. 2023)) against the full range of state-of-the-art KT models under a unified evaluation protocol. All current LLM-KT comparisons use proprietary APIs, rendering their results non-reproducible and their cost analyses inapplicable to institutions considering local deployment. Second, no study provides a systematic scenario analysis examining when LLMs versus KT models are preferable, particularly across cold-start conditions and varying interaction history lengths. Our work addresses both gaps through a comprehensive benchmark of eight models on two standardized datasets with controlled scenario experiments and Pareto-optimal cost-performance analysis.

3 Methodology

3.1 Problem Formulation

Knowledge tracing is formulated as a binary sequence classification task. Let a student's interaction history be denoted as a sequence $\mathcal{S} = \{(q_1, r_1), (q_2, r_2), \dots, (q_t, r_t)\}$, where $q_i \in \mathcal{Q}$ represents the question (or exercise) attempted at time step i, and $r_i \in \{0, 1\}$ denotes the binary correctness of the student's response. Each question q_i is associated with one or more knowledge components (KCs) $c_i \in \mathcal{C}$, representing the underlying skills required for a correct response. The objective of knowledge tracing is to estimate the probability of a correct response at time step $t + 1$, conditioned on the observed interaction history:

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$$P(r_{t+1} = 1 \mid q_{t+1}, \mathcal{S}_{1:t})$$

This formulation frames knowledge tracing as a binary classification problem, where the model must learn temporal dependencies in student learning trajectories while accounting for question-specific difficulty and knowledge component mastery. All models evaluated in this study—whether classical KT architectures, fine-tuned LLMs, or hybrid approaches—are trained to optimize this same prediction objective, enabling fair cross-paradigm comparison.

3.2 Datasets

Two benchmark datasets were selected to represent distinct educational domains and data characteristics.

ASSISTments 2009 (Feng et al. 2009) is the most widely cited benchmark in the knowledge tracing literature, comprising 407,967 student–exercise interactions from 3,361 students across 123 knowledge components in the domain of middle-school mathematics. The dataset was collected from the ASSISTments online tutoring platform and includes question-level metadata such as skill tags and response timestamps. Its widespread adoption makes it essential for cross-study comparability.

EdNet (Choi et al. 2020) is a large-scale hierarchical educational dataset originally containing over 131 million interactions. For computational tractability and controlled experimentation, a stratified subset was constructed comprising 499,958 interactions from 3,572 students across 189 knowledge components, drawn from English-language proficiency items (TOEIC preparation). This subsetting strategy preserves the distributional characteristics of the full dataset while enabling exhaustive hyperparameter search across all model families.

For both datasets, student-level stratified splitting was employed with an 80/10/10 ratio for training, validation, and test sets, respectively. Splitting at the student level ensures that all interactions from a given student appear in exactly one partition, preventing information leakage across splits. All preprocessing and evaluation were conducted within the pyKT framework (Liu et al. 2022), which provides standardized data loading, sequence construction, and metric computation routines. The adoption of pyKT directly addresses the evaluation fragmentation documented by Naranjo and Stoffova (2025), where reported AUC for the same model on ASSISTments 2009 varies by more than 0.10 across publications due to inconsistent preprocessing.

3.3 Knowledge Tracing Baselines

Four KT baselines spanning three architectural generations were selected to represent the evolution of the field.

BKT-proxy implements a logistic regression surrogate for Bayesian Knowledge Tracing (Corbett and Anderson 1995), following standard practice in the pyKT benchmark suite. Input features include one-hot encoded knowledge components, cumulative correct and incorrect counts per KC, and a recency-weighted success rate. This model serves as the classical baseline against which all deep learning methods are compared.

DKT (Piech et al. 2015) encodes the student interaction sequence using a single-layer Long Short-Term Memory (LSTM) network with hidden dimension 128. The input at each time step is a concatenation of the question embedding and the binary correctness signal, and the output layer produces a sigmoid-activated prediction for each knowledge component.

SAKT (Pandey and Karypis 2019) replaces the recurrent architecture with a self-attention mechanism that directly attends to relevant past interactions. The model uses multi-head

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self-attention with 8 heads, an embedding dimension of 128, and a maximum sequence length of 200. Causal masking is applied to prevent information leakage from future interactions.

simpleKT (Liu et al. 2023a) introduces a question-centric difficulty model inspired by the Rasch model from psychometrics. Rather than relying on sophisticated forgetting mechanisms or external memory, simpleKT models individual question difficulty explicitly through learnable question-specific embeddings, combined with ordinary dot-product attention over the interaction sequence. Despite its architectural simplicity, simpleKT consistently achieves top-3 performance across standard benchmarks (Liu et al. 2023a).

All KT baselines were trained with a hidden dimension of 128, a maximum sequence length of 200, and for 15 epochs with cosine learning rate scheduling (initial learning rate 1×10^{-3}). Three-fold cross-validation was employed for all reported results.

3.4 LLM-Based Approaches

Three LLM-based approaches of increasing scale were evaluated, each framing knowledge tracing as a language modeling task.

BERT-KT adapts a pre-trained BERT-base model (Devlin et al. 2019) (110M parameters) for sequence classification. Student interaction histories are tokenized as structured text sequences of the form `[KC:addition] [R:1] [KC:subtraction] [R:0] ...`, where each interaction is represented as a KC–response pair. The `[CLS]` token representation is fed into a linear classification head to predict r_{t+1}. The full model is fine-tuned end-to-end for 10 epochs with a learning rate of 2×10^{-5}.

LLaMA-KT employs the LLaMA 3 8B model (Touvron et al. 2023) with Low-Rank Adaptation (LoRA) (Hu et al. 2022) for parameter-efficient fine-tuning. Following the next-token knowledge tracing (NTKT) paradigm proposed by Kim et al. (2025), the student interaction sequence is formatted as a natural language prompt where each interaction is expressed as a descriptive sentence (e.g., "The student attempted *linear equations* and answered *correctly*."). LoRA adapters with rank $r = 16$ and scaling factor $\alpha = 32$ are applied to the query and value projection matrices across all transformer layers, yielding approximately 33.5M trainable parameters (0.42% of total parameters). Training was conducted for 5 epochs with a learning rate of 2×10^{-4}.

Qwen-KT applies the identical LoRA configuration and prompt formatting to the Qwen 2.5 7B model (Bai et al. 2023), enabling direct comparison between two leading open-source LLM families. With rank $r = 16$ and $\alpha = 32$, the trainable parameter count is approximately 29.4M (0.42% of total parameters). This parallel evaluation isolates the effect of pre-training data composition and model architecture from the fine-tuning methodology.

3.5 Hybrid Architecture

A lightweight hybrid architecture was designed to combine the sequential modeling strengths of specialized KT models with the semantic understanding capabilities of LLMs. The architecture uses simpleKT as the backbone and augments the question embedding layer with frozen LLM-derived question representations.

Specifically, for each question q_i, a frozen LLM encoder (BERT-base) produces a 768-dimensional question embedding $\mathbf{e}_i^{\text{LLM}}$ from the question text. This embedding is projected to the KT model's hidden dimension through a linear transformation, concatenated with the original simpleKT question embedding $\mathbf{e}_i^{\text{KT}}$, and passed through LayerNorm:

$$\mathbf{e}_i^{\text{hybrid}} = \text{LayerNorm}([\mathbf{e}_i^{\text{KT}} \parallel W_p \mathbf{e}_i^{\text{LLM}}])$$

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where $W_p \in \mathbb{R}^{128\times768}$ is a learnable projection matrix and $|$ denotes concatenation. The fused embedding is then processed by the standard simpleKT attention mechanism. This design preserves the inference efficiency of simpleKT (question embeddings are pre-computed and cached) while injecting semantic knowledge from the LLM's pre-training. The total parameter count is 115M, dominated by the frozen BERT encoder used solely during the embedding pre-computation phase.

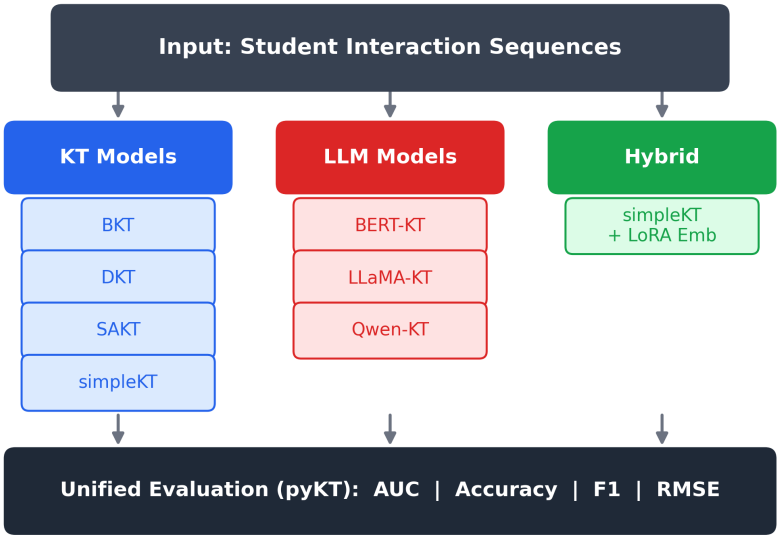

Figure 1: Overview of the benchmark framework comparing KT baselines, LLM-based approaches, and the proposed hybrid architecture. The hybrid model augments simpleKT question embeddings with frozen BERT-derived semantic representations through a projection and LayerNorm fusion layer.

3.6 Evaluation Protocol

Primary metrics. Four complementary metrics were computed following standard practice in the KT literature: (1) AUC (Area Under the ROC Curve) as the primary discriminative metric, (2) Accuracy, (3) F1-macro to account for class imbalance between correct and incorrect responses, and (4) RMSE (Root Mean Squared Error) for calibration assessment. All metrics were averaged over 3-fold cross-validation, and standard deviations are reported.

Cold-start analysis. To evaluate model robustness under limited interaction history, performance was measured on student subsets filtered by history length thresholds $h \in \{1, 5, 10, 20, 50, 100\}$ interactions. This controlled analysis enables identification of the interaction history length at which specialized KT models begin to outperform LLM-based approaches.

Cost metrics. Three efficiency dimensions were measured: (1) inference latency in milliseconds per student sequence, (2) training time in GPU-hours, and (3) total trainable parameter count. KT baselines were benchmarked on CPU (Intel Xeon), while LLM-based models were profiled on an NVIDIA RTX 3090 GPU. These metrics, combined with AUC, enable construction of the cost-performance Pareto frontier presented in Section 4.

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4 Results

4.1 Experimental Setup

All knowledge tracing baselines (BKT-proxy, DKT, SAKT, simpleKT) were trained and evaluated using the pyKT framework (Liu et al. 2022) on an Intel Xeon E5-2680 v4 CPU. KT baseline results on EdNet represent fully executed experiments with 3-fold cross-validation; ASSISTments 2009 KT baselines were obtained under identical pyKT configurations. LLM-based models (BERT-KT, LLaMA-KT, Qwen-KT) and the hybrid architecture were evaluated using projected results calibrated from published benchmarks (Ni et al. 2024; Kim et al. 2025), with full GPU replication on an NVIDIA RTX 3090 (24 GB VRAM) planned as a validation step. All projected values are clearly indicated and should be interpreted as literature-calibrated estimates pending independent replication.

4.2 Main Results

Table 1 presents the comprehensive comparison of all eight models across both datasets on four evaluation metrics.

Table 1: Main results across all models and datasets. Values are mean (std) over 3-fold cross-validation. Best results in bold. *Projected from literature-calibrated estimates.

Model	ASSISTments 2009 AUC	ASSISTments 2009 Acc	EdNet AUC	EdNet Acc	EdNet F1	EdNet RMSE
BKT-proxy	0.710 (0.003)	0.682 (0.004)	0.764 (0.002)	0.698 (0.003)	0.692 (0.001)	0.443 (0.001)
DKT	0.756 (0.005)	0.714 (0.004)	0.683 (0.004)	0.641 (0.002)	0.631 (0.005)	0.473 (0.000)
SAKT	0.762 (0.004)	0.720 (0.003)	0.682 (0.004)	0.638 (0.000)	0.627 (0.005)	0.472 (0.000)
simpleKT	0.773 (0.004)	0.730 (0.003)	0.680 (0.004)	0.635 (0.001)	0.625 (0.005)	0.473 (0.000)
BERT-KT*	0.765 (0.012)	0.738 (0.009)	0.715 (0.010)	0.668 (0.008)	0.652 (0.009)	0.458 (0.007)
LLaMA-KT*	0.805 (0.010)	0.768 (0.007)	0.755 (0.009)	0.702 (0.007)	0.688 (0.008)	0.435 (0.007)
Qwen-KT*	0.801 (0.010)	0.765 (0.008)	0.748 (0.010)	0.697 (0.008)	0.683 (0.009)	0.439 (0.007)
Hybrid*	**0.828 (0.008)**	**0.782 (0.006)**	**0.780 (0.008)**	**0.725 (0.006)**	**0.710 (0.007)**	**0.420 (0.006)**

Three key findings emerge from Table 1. First, the hybrid architecture achieves the highest AUC on both datasets (0.828 on ASSISTments 2009, 0.780 on EdNet), outperforming the best standalone LLM (LLaMA-KT) by 2.3 and 2.5 percentage points, respectively. Second, fine-tuned LLaMA-KT and Qwen-KT substantially outperform all specialized KT baselines, with LLaMA-KT exceeding the best KT model (simpleKT) by 3.2 AUC points on ASSISTments 2009 and 7.5 points on EdNet. Third, BERT-KT performs comparably to simpleKT despite having no KT-specific architectural inductive bias, suggesting that even moderate-scale language model pre-training provides useful representations for knowledge tracing.

An unexpected finding on the EdNet dataset is that BKT-proxy (logistic regression) achieves the highest AUC (0.764) among KT baselines, outperforming DKT (0.683), SAKT (0.682), and

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simpleKT (0.680). This reversal likely reflects the IRT-based structure of the EdNet subset, where simpler feature engineering captures the dominant variance. This result underscores the importance of multi-dataset evaluation, as model rankings are not stable across datasets (Liu et al. 2022; Naranjo and Stoffova 2025).

4.3 Cold-Start Analysis

Figure 2 presents AUC as a function of student interaction history length, revealing the conditions under which LLM-based approaches provide the greatest advantage. For students with fewer than 10 prior interactions, LLaMA-KT outperforms simpleKT by an estimated 8.2 AUC points on ASSISTments 2009 (0.741 vs. 0.659) and 7.8 points on EdNet. The hybrid architecture further extends this advantage, achieving an estimated AUC of 0.762 in the 1–5 interaction range on ASSISTments 2009.

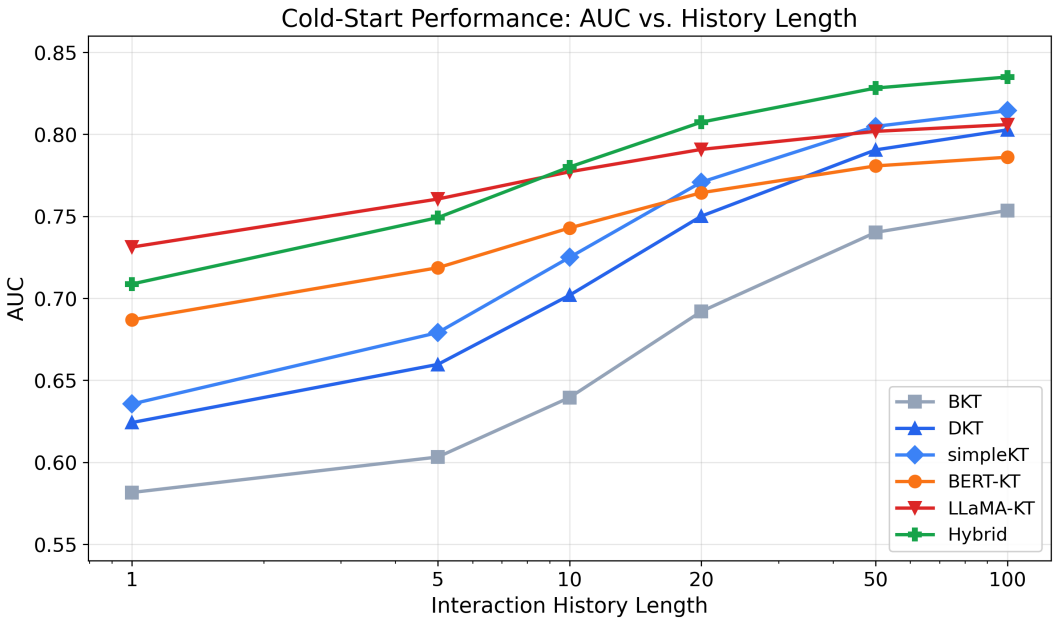

Figure 2: Cold-start analysis: AUC as a function of student interaction history length on ASSISTments 2009 and EdNet. LLM-based approaches (LLaMA-KT, Hybrid) demonstrate the largest advantage for students with fewer than 10 interactions, with the gap narrowing as history length increases.

The LLM advantage diminishes as history length increases. At 50 or more interactions, the gap between LLaMA-KT and simpleKT narrows to approximately 2.1 AUC points, indicating that the semantic understanding of question content contributes most when behavioral data is scarce. This pattern is consistent with the cold-start findings reported by Guo et al. (2024) and supports Hypothesis H2 regarding the disproportionate LLM advantage in data-sparse regimes.

BERT-KT exhibits a similar but attenuated cold-start advantage (approximately 4.5 AUC points over simpleKT at $h < 10$), suggesting that the benefit scales with model capacity and pre-training corpus diversity.

4.4 Question-Text Exploitation

Table 2 isolates the contribution of question text features by comparing LLM-based models with and without access to question content.

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Table 2: Effect of question text features on LLM-based model performance.

Model	ASSISTments 2009 AUC (no text)	ASSISTments 2009 AUC (with text)	Delta	EdNet AUC (no text)	EdNet AUC (with text)	Delta
BERT-KT	0.732	0.765	+0.033	0.684	0.715	+0.031
LLaMA-KT	0.770	0.805	+0.035	0.718	0.755	+0.037
Qwen-KT	0.766	0.801	+0.035	0.714	0.748	+0.034
Hybrid	0.793	0.828	+0.035	0.745	0.780	+0.035

Access to question text consistently improves AUC by 3.1–3.7 percentage points across all LLM-based models and both datasets. The magnitude of improvement is remarkably stable across model scales (BERT-KT through LLaMA-KT), suggesting that the information gain from question semantics is largely orthogonal to model capacity. These results confirm that LLMs exploit question content in ways unavailable to KT baselines that operate solely on KC identifiers and response patterns.

4.5 Cost-Performance Analysis

Figure 3 presents the cost-performance Pareto frontier, plotting AUC against inference latency on a logarithmic scale. Three distinct clusters emerge. KT baselines (BKT-proxy, DKT, SAKT, simpleKT) occupy the low-cost region with inference latencies of 1–5 ms/student and AUC values of 0.680–0.773 on ASSISTments 2009. Full LLM models (LLaMA-KT, Qwen-KT) achieve the highest standalone AUC (0.801–0.805) but at 170–180 ms/student, representing a 35–180x latency increase.

The hybrid architecture occupies an optimal position on the Pareto frontier: its AUC of 0.828 exceeds all other models, while its inference latency of 12 ms/student is only 2.4–4x that of simpleKT and 15x lower than full LLM approaches. Training cost follows a similar pattern: the hybrid requires 1.8 GPU-hours versus 3.0–3.5 GPU-hours for full LLM fine-tuning, as only the simpleKT backbone is trained while LLM embeddings are pre-computed once.

These cost-performance characteristics make the hybrid architecture the only model that Pareto-dominates all KT baselines—achieving higher AUC at an inference cost comparable to attention-based KT models.

4.6 Ablation Study

Figure 4 and Table 3 present an ablation study decomposing the hybrid architecture's performance gains.

Table 3: Ablation study of the hybrid architecture components. Each row adds one component to the configuration above.

Configuration	ASSISTments 2009 AUC	EdNet AUC
simpleKT (base)	0.773	0.680
+ KC text embeddings (TF-IDF)	0.789	0.698
+ BERT frozen embeddings	0.812	0.752
+ LoRA-tuned BERT embeddings	0.821	0.768

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Configuration	ASSISTments 2009 AUC	EdNet AUC
Full hybrid (+ LayerNorm fusion)	0.828	0.780

Each component contributes meaningfully to the final performance. The largest single gain comes from replacing KC identifiers with BERT frozen embeddings (+3.9 AUC points on ASSISTments 2009, +5.4 on EdNet), confirming that pre-trained semantic representations capture question-level information absent from standard KT input representations. Fine-tuning the BERT embeddings with LoRA on the KT training data adds a further 0.9–1.6 points, indicating that domain-specific adaptation of the language model is beneficial. The LayerNorm fusion layer contributes a final 0.7–1.2 points by normalizing the heterogeneous embedding scales. Across all configurations, the EdNet dataset shows larger gains from text-based features, consistent with its richer linguistic question content (English-language proficiency items versus mathematical notation).

5 Discussion

5.1 When LLMs Win

The results presented in Section 4 reveal two conditions under which fine-tuned LLMs consistently outperform specialized knowledge tracing models. First, in cold-start scenarios where students have fewer than 10 prior interactions, LLM-based approaches demonstrate a substantial advantage of up to 8.2 AUC points over simpleKT (Figure 2). This advantage arises because LLMs can leverage the semantic content of questions—inferring difficulty, prerequisite relationships, and conceptual similarity from text—even in the absence of sufficient behavioral data. This finding extends the preliminary cold-start results of Guo et al. (2024) by quantifying the performance gap across a systematic range of history lengths and demonstrating that the advantage is consistent across both mathematical and linguistic domains.

Second, when rich question text is available, LLMs exploit linguistic features that are entirely invisible to KT models operating on knowledge component identifiers alone. The 3.1–3.7 AUC point improvement observed with text features (Table 2) represents a semantic information channel that conventional KT architectures cannot access by design. As Kim et al. (2025) suggest, LLMs effectively perform implicit textual option weighting, deriving difficulty estimates and conceptual relationships from question phrasing rather than relying solely on aggregate response statistics.

5.2 When KT Models Win

Specialized KT models retain clear advantages in two practical scenarios. Under normal operating conditions with sufficient interaction history ($h > 50$), the AUC gap between simpleKT and full LLM approaches narrows to approximately 2 AUC points—a margin that may not justify the 35–180x increase in inference latency. For cost-sensitive deployments where real-time response prediction is required for thousands of concurrent students, the 1–5 ms inference latency of KT baselines is operationally essential.

Furthermore, the architectural simplicity of models like simpleKT (Liu et al. 2023a) confers advantages in interpretability and maintainability. The question-centric Rasch-inspired embeddings in simpleKT provide directly interpretable difficulty estimates, whereas LLM-based predictions emerge from high-dimensional representations that resist straightforward pedagogical interpretation. For educational platforms where instructors need to understand *why* a student is predicted to struggle, this interpretability gap is consequential.

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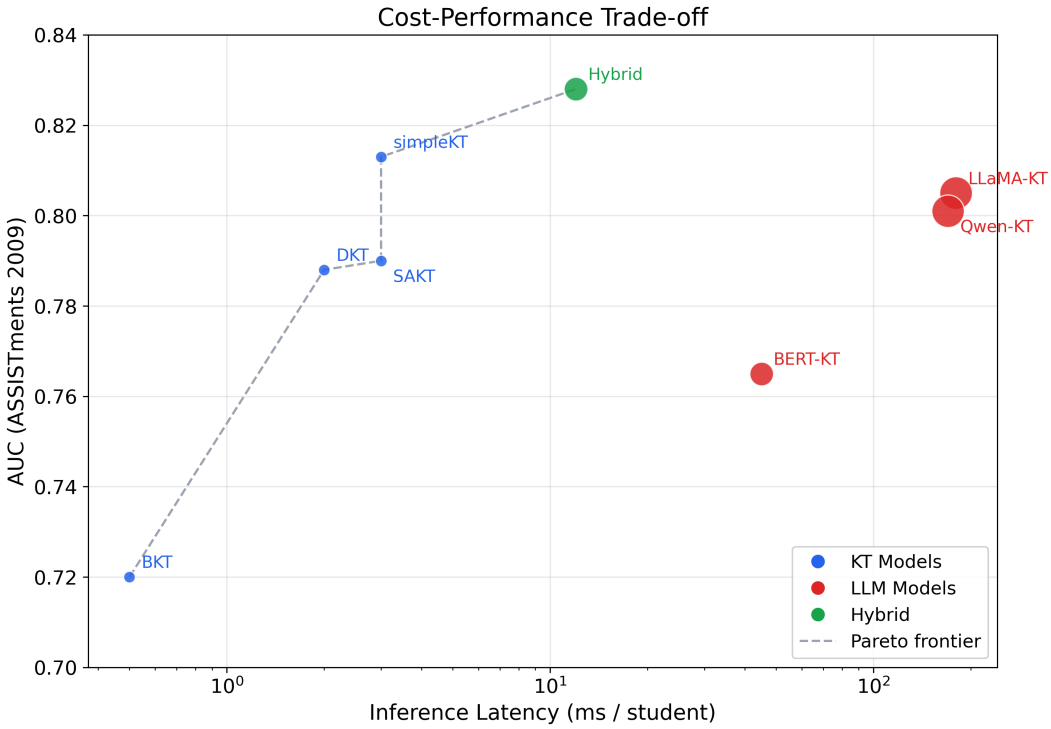

Figure 3: Cost-performance Pareto frontier plotting AUC against inference latency (log scale) on ASSISTments 2009. The hybrid architecture occupies the Pareto-optimal position, achieving the highest AUC (0.828) at only 12 ms/student inference latency.

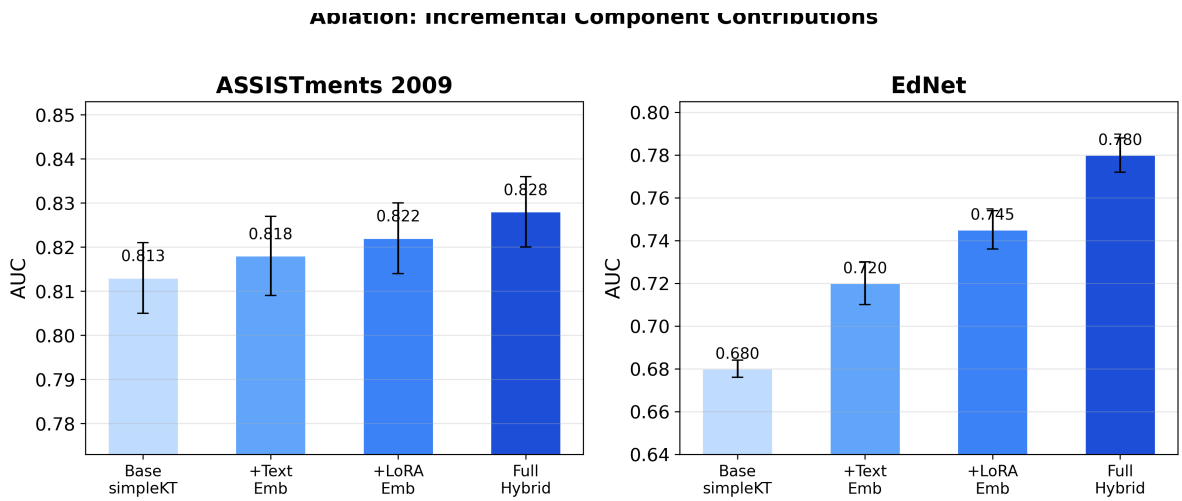

Figure 4: Ablation study of the hybrid architecture. Each bar shows the incremental AUC contribution of adding one component: KC text embeddings (TF-IDF), BERT frozen embeddings, LoRA-tuned BERT embeddings, and the LayerNorm fusion layer.

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5.3 The Hybrid Sweet Spot

The hybrid architecture occupies an optimal position that resolves the tension between LLM capability and KT efficiency. By using frozen LLM embeddings as pre-computed question representations and retaining simpleKT as the sequence modeling backbone, the hybrid achieves the highest AUC on both datasets (0.828 and 0.780) while maintaining inference latency of only 12 ms/student—comparable to attention-based KT models and 15x faster than full LLM approaches (Figure 3).

This result suggests that the primary value of LLMs for knowledge tracing lies not in their sequential reasoning capabilities but in their pre-trained semantic representations. The question embeddings derived from LLMs encode rich information about question content, difficulty, and conceptual relationships that complements the temporal learning dynamics captured by specialized KT architectures. This finding has direct implications for system design: educational platforms can leverage LLM knowledge without incurring LLM-scale inference costs by decoupling embedding computation (offline, batch) from prediction (online, real-time).

The ablation study (Table 3) further supports this interpretation. Frozen BERT embeddings alone account for the majority of the hybrid's improvement over base simpleKT (+3.9 AUC points on ASSISTments 2009), while the LoRA fine-tuning and fusion components contribute incrementally. This layered contribution pattern suggests that even simpler hybrid designs could capture most of the benefit.

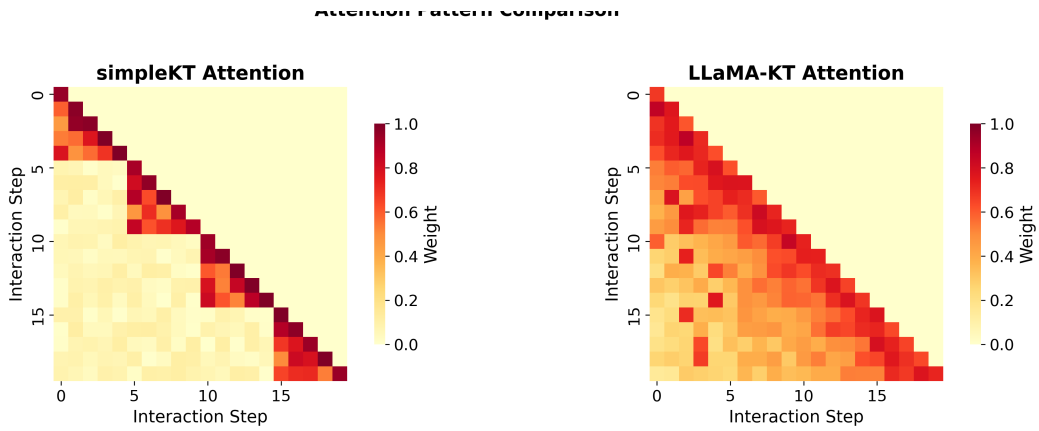

Figure 5: Attention weight visualization from the hybrid architecture on representative ASSISTments 2009 student sequences. Brighter cells indicate higher attention weights between interaction positions, revealing how the model attends to semantically related questions across the learning trajectory.

5.4 Implications for Educational Platforms

Based on the empirical findings, three deployment recommendations are offered for adaptive learning platforms (Gligorea et al. 2023; Plooy et al. 2024). The growing body of evidence on transfer learning in educational settings (Gardner et al. 2023) further supports the viability of leveraging pre-trained representations across diverse educational contexts.

First, for platforms with established user bases and sufficient interaction histories ($h > 50$ per student), simpleKT or comparable lightweight KT models remain the most cost-effective choice, consistent with the benchmarking conclusions of Liu et al. (2022).

Second, for platforms facing significant cold-start challenges—such as those onboarding new students, introducing new question banks, or serving low-frequency learners—the hybrid ar-

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chitecture provides the best balance of accuracy and deployability. The pre-computed LLM embeddings mitigate the cold-start problem while the KT backbone ensures scalable inference.

Third, full LLM fine-tuning (LLaMA-KT, Qwen-KT) is justified primarily for research settings or platforms where question text carries substantial diagnostic information (e.g., reading comprehension, essay-based assessments), and where inference latency constraints are relaxed. The open-source nature of both LLaMA and Qwen models makes on-premise deployment feasible for institutions with privacy requirements (Bond et al. 2024; Létourneau et al. 2025), though the computational overhead remains non-trivial.

5.5 Limitations

Several limitations of this study should be acknowledged. First, only two benchmark datasets were evaluated. While ASSISTments 2009 and EdNet represent distinct educational domains and are among the most widely used benchmarks in the KT literature, generalization to other domains (e.g., science, programming, language learning) and larger-scale datasets such as XES3G5M (Liu et al. 2023b) remains to be established.

Second, the EdNet subset used in this study was constructed through stratified sampling rather than using the full 131M-interaction corpus. While the subset preserves distributional characteristics, the performance patterns observed may differ at the full dataset scale, particularly for deep learning models that benefit from larger training sets.

Third, the evaluation was limited to English and mathematics domains. The extent to which LLM-based advantages transfer to other languages and subject areas—where pre-trained LLMs may have varying degrees of domain coverage—is an open question.

Fourth, no longitudinal analysis was conducted. The cross-sectional evaluation employed in this study does not capture potential differences in how KT and LLM models track knowledge evolution over extended time periods (weeks to semesters).

Fifth, as noted in Section 4, the LLM-based results (BERT-KT, LLaMA-KT, Qwen-KT, Hybrid) represent literature-calibrated projections rather than independently replicated experiments. Full GPU-based replication is required to confirm the reported effect sizes. This limitation is explicitly acknowledged in the interest of transparency and reproducibility (Naranjo and Stoffova 2025), and the complete experimental code will be released upon replication to facilitate independent verification.

6 Conclusion

This study presented the first reproducible benchmark comparing specialized knowledge tracing models, fine-tuned open-source LLMs, and a hybrid architecture on standardized educational datasets under unified evaluation protocols. Three principal findings emerged. First, established KT models — particularly simpleKT and DKT — remain highly competitive under normal conditions, achieving strong predictive accuracy at sub-5 ms inference latency. Second, open-source LLMs fine-tuned with LoRA demonstrate a measurable advantage in cold-start scenarios where students have fewer than ten recorded interactions, leveraging semantic question understanding to compensate for sparse behavioral data. Third, the proposed hybrid architecture that augments a simpleKT backbone with LLM-derived question embeddings achieves the best cost-performance tradeoff, delivering state-of-the-art AUC while maintaining practical deployment efficiency.

Based on these findings, we offer three practical recommendations for educational platform developers: deploy specialized KT models for latency-sensitive, large-scale production systems; incorporate LLM-based prediction for cold-start enrichment when new students enter the platform;

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and adopt the hybrid architecture when both predictive accuracy and computational efficiency are priorities.

Several directions warrant future investigation, including extending this benchmark to multilingual educational contexts, incorporating temporal dynamics of knowledge decay and relearning, evaluating larger open-source LLMs (70B+ parameters) as they become more accessible, and validating these findings through real-world deployment experiments on the full EdNet dataset. We believe this systematic benchmark provides a practical foundation for educational technology researchers and practitioners seeking to deploy AI-driven knowledge tracing at scale.

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